

Consensus statement: updated recommendations for the interdisciplinary management of people living with HIV in Romania

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Abstract

Owing to significant advances in HIV treatment and the resultant increase in life expectancy, the number of aging individuals living with HIV and associated comorbidities continues to rise. Consequently, the management of people living with HIV is no longer solely the responsibility of infectious disease or HIV specialists, but requires an integrated and multidisciplinary approach that addresses the prevention, as well as the monitoring and treatment needs of associated conditions. The care of people living with HIV in Romania is largely aligned with international guidelines, particularly those of the European AIDS Clinical Society (EACS). However, guideline implementation requires adaptation to local clinical realities and collaboration across medical specialties. In response to this need, a team of experts in infectious diseases, cardiology, nephrology, diabetes, metabolic disorders, and clinical psychology, convened to develop a national consensus for the interdisciplinary management of people living with HIV. The consensus provides clear and practical recommendations addressed to both infectious disease specialists and healthcare providers from other specialties involved in the care of people living with HIV. Its goal is to offer a unified, up-to-date, and applicable framework to support patient-centered care, facilitate interdisciplinary collaboration, and contribute to improving the quality of life of people living with HIV in Romania.

Keywords HIV, people living with HIV, comorbidities, interdisciplinary, prevention, treatment.

Introduction

Human immunodeficiency virus (HIV) infection continues to represent a major global public health challenge, with over 40 million people currently living with HIV worldwide.¹ Owing to significant advances in HIV treatment and the resultant increase in life expectancy, the

number of aging individuals living with HIV and associated comorbidities continues to rise.^{2,4} By 2030, it is estimated that nearly half of all people with HIV will be living with multiple chronic conditions.⁴

In Romania, more than 18,000 individuals are living with HIV according to the latest

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estimates, with over 800 new cases diagnosed in 2024.⁵ Many of these individuals require medical care that extends beyond the field of infectious diseases (ID). However, despite significant improvement over the past few years, access to such multidisciplinary care sometimes remains limited, on the one hand due to persistent systemic HIV-related stigma,⁶ and on the other hand due to insufficient experience among non-ID specialists in managing people living with HIV.

This consensus statement outlines updated principles regarding interdisciplinary care of people living with HIV in Romania, based on international guidelines and expert opinion. The consensus also provides a framework for recommendations tailored to several medical specialties that often are or should be involved in the care of people living with HIV.

Methods

In December 2024, a multidisciplinary group of specialists in ID, cardiology, diabetes and metabolic diseases, nephrology and clinical psychology created a consensus working group for the review and update of practice algorithms for multidisciplinary care of people living with HIV in Romania. Each expert was asked to perform a

preliminary review of all relevant national and international guidelines applicable in their respective medical specialty, as well as the recommendations from the 2024 version 12.1 of the European AIDS Clinical Society (EACS) Guidelines.⁷ Following an initial in-person meeting, the experts drafted a consensus statement that complements and updates a previous consensus document published in 2019 for Romania.⁸ The final draft of the consensus recommendations were reviewed and validated by the heads of all nine HIV regional centers in Romania (Bucharest Matei Balș, Bucharest Victor Babeș, Brașov, Cluj-Napoca, Constanța, Craiova, Iași, Târgu-Mureș, Timișoara), to ensure national consistency, alignment and applicability.

Testing, diagnosis and assessment

HIV infection can be identified through routine screening, or through targeted testing in individuals at increased risk or in patients who present clinical signs and symptoms suggestive of HIV indicator conditions. Regardless of the context, confirmed cases of HIV infection should be promptly referred to an ID specialist for comprehensive evaluation, screening for opportunistic infections, and rapid initiation of antiretroviral therapy (ART). This process is most often accompanied by psychological counseling, which represents an essential component of early care. All regional HIV centers in Romania are equipped to provide both appropriate medical management and psychological support to patients living with HIV.

HIV testing should be routinely offered to all individuals as part of regular medical care and periodic health assessments. Certain populations at increased risk of acquiring HIV (men who have sex with men, persons with multiple sexual partners or high-risk sexual contacts, chemsex users, recipients of pre-exposure prophylaxis (PrEP), individuals diagnosed with sexually transmitted infections (STIs) or acute viral hepatitis, and people who inject drugs and share needles or syringes) should be screened more frequently, for example every 3 to 6 months, based on individual risk profiles.⁹

Acute HIV infection is often asymptomatic or presents as a non-specific, mononucleosis-like

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syndrome. It is then followed by years of silent evolution until HIV indicator conditions first appear. Consequently, HIV testing can frequently be overlooked. Therefore, obtaining a detailed medical history, including risk factors for HIV acquisition, can be instrumental in facilitating early diagnosis and timely initiation of ART, and reducing the risk of onward transmission.

At the time of diagnosis and regularly further on, a series of assessments and exams are needed for optimal overall management of people living with HIV (Figures 1 and 2).⁷ In Romania, all people living with HIV have access to ART and specialized HIV care and counselling, coordinated by an ID specialist. Referrals to other medical specialists are often needed and interdisciplinary collaboration represents the mainstay of contemporary medical care.

Prevention and management of cardiovascular disease

HIV infection is associated with a marked risk of cardiovascular comorbidities, related to infection itself, treatment history, and chronic inflammation.^{10,11} The presence of HIV infection doubles the pooled risk of heart failure and atherosclerotic cardiovascular disease (ASCVD) events, even after adjustment for several clinical factors, including age, blood pressure values, cholesterol levels and presence of diabetes.¹² Importantly, even in persons with well controlled HIV infection, with suppressed HIV viral load and high CD4 cell counts, a residual cardiovascular (CV) risk remains.^{7,13-15}

CV risk assessment should be performed at least annually in all people living with HIV. According to the updated EACS guidelines,⁷ SCORE2 is the recommended tool for CV risk assessment in people aged 40 to less than 70 years without ASCVD, diabetes, chronic kidney disease (CKD), or familial hypercholesterolemia. Still, in people living with HIV, this tool may underestimate actual risk, making clinical judgment and a comprehensive patient assessment particularly important.

Emphasis should be placed on recognizing and addressing traditional modifiable CV risk factors such as smoking, obesity, sedentary lifestyle, hypertension, dyslipidemia, or

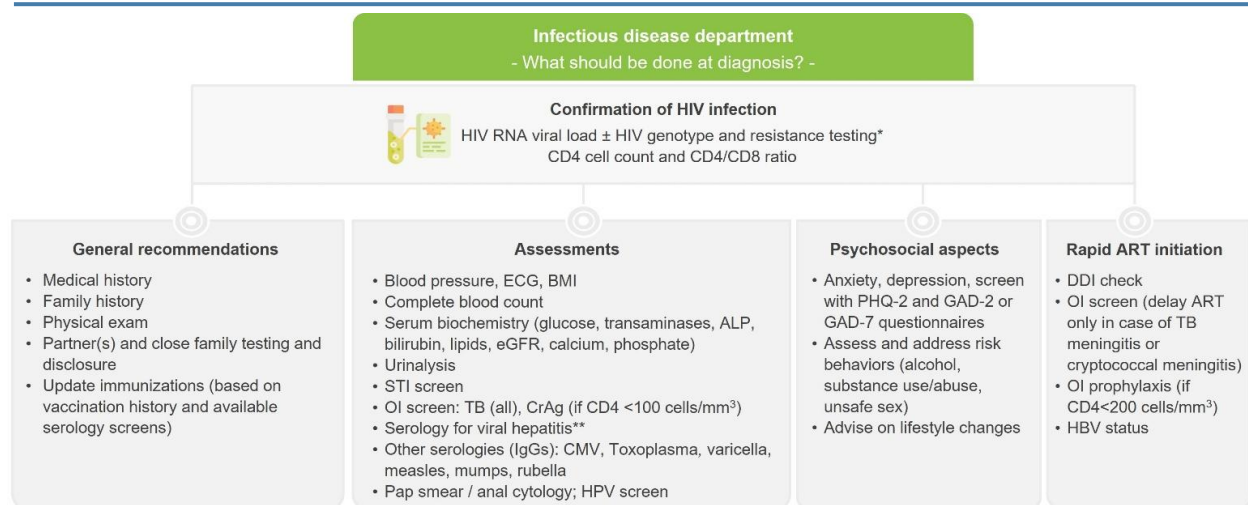
uncontrolled diabetes mellitus. Furthermore, non-traditional risk factors for CV disease are increasingly being described, including depression and alcohol abuse.¹⁶ It is essential to discuss all these items with the patients, and to propose comprehensive mitigation measures for each of the identified risk factors.

A structured approach to CV risk assessment and management in people living with HIV involves close collaboration between ID specialists and cardiologists. During clinical consultations, a minimum set of explorations should include blood pressure measurement, lipid profile determination, and body mass index (BMI) calculation. Supplementary investigations may include electrocardiogram (ECG), carotid Doppler ultrasound, or echocardiography, as appropriate. Figure 3 summarizes practical recommendations regarding blood pressure control, lipid management, and the evaluation of pulmonary hypertension. These include risk stratification, lifestyle interventions, pharmacologic treatment tailored to HIV-related considerations, and timely referral for specialized cardiology and/or pneumology evaluation.

Prevention of major adverse cardiovascular events is an important goal in the management of patients living with HIV. Measures taken for prevention include addressing the risk factors described above, obtaining good control of comorbidities such as hypertension or diabetes mellitus, as well as targeted ART-related modifications such as switching away from regimens containing abacavir or protease inhibitors. Prevention strategies also include moderate-intensity or high-intensity statin treatment, based on SCORE2 risk assessment. Among statins, pitavastatin showed significant risk reduction in people living with HIV and is included as preferred option in EACS guidelines.^{7,17} At this moment, pitavastatin is not available in Romania, similarly to many other countries, and therefore other statin options should be used, such as atorvastatin or rosuvastatin, with dose titration and monitoring.

Kidney disease

People living with HIV have a higher risk of kidney disease. Renal involvement in HIV is



*In the majority of cases, there is no need to wait for resistance testing results prior to ART initiation. If resistance testing cannot be performed, at a minimum, a baseline sample should be collected and stored.

**Triple panel test for HBV: HBsAg, HBcAb (total or IgG), HBsAb (if HBsAg positive, anti-HDV); anti-HCV (if positive, HCV RNA); HAV IgG (if negative, recommend HAV vaccination).

ALP – alkaline phosphatase; ART – antiretroviral therapy; BMI – body mass index; CMV – cytomegalovirus; CrAg – serum cryptococcal antigen; DDI – drug-drug interactions; ECG – electrocardiogram; eGFR – estimated glomerular filtration rate; GAD-2 – generalized anxiety disorder 2-item; GAD-7 – generalized anxiety disorder 7-item; HBV – hepatitis B virus; HIV RNA – human immunodeficiency virus ribonucleic acid; HPV – human papillomavirus; IgGs – immunoglobulins G; OI – opportunistic infections; PHQ-2 – patient health questionnaire-2; STI – sexually transmitted infections; TB – tuberculosis.

Figure 1. General overview of initial ID assessments following HIV diagnosis

Clinical scenario	HBsAg	HBcAb (IgG or total)	HBsAb	Recommended action: Further investigations	Recommended action: Treatment
Current HBV infection	Positive (+)	Positive (+)	Negative (-)	HBV DNA; anti-HDV (if positive: HDV RNA); monitor liver fibrosis, screen for HCC (regular AFP and liver US)	TDF/TAF-containing ART
Prior HBV vaccination	Negative (-)	Negative (-)	Positive (+)	N/A	N/A
Prior HBV infection or Occult HBV infection	Negative (-)	Positive (+)	Negative or Positive (-) / (+)	HBV DNA	TDF/TAF-containing ART
Prior resolved HBV infection	Negative (-)	Positive (+)	Positive (+)	Repeat triple panel yearly or sooner in case of immunosuppression	TDF/TAF containing ART in case of current or future immunosuppression
Negative HBV markers	Negative (-)	Negative (-)	Negative (-)	N/A Vaccinate and repeat triple panel after vaccination, to check response	HBV vaccination (3-dose regimen) In case of non-response: • repeat 3-dose regimen; • use HepB-CpG vaccine, if available; • TDF/TAF-containing ART

Positive HBsAb defined as >10 mIU/mL. Anti-HDV – hepatitis delta virus antibodies; AFP – alpha-fetoprotein; ART – antiretroviral therapy; HBV – hepatitis B virus; HBsAg – HBV surface antigen; HBcAb – HBV core antibodies; HBsAb – HBV surface antibodies; HBV DNA – HBV deoxyribonucleic acid; HCC – hepatocellular carcinoma; HDV RNA – hepatitis delta virus ribonucleic acid; HepB-CpG – hepatitis B vaccine with a cytosine phosphoguanine adjuvant; US – ultrasound.

Figure 2. Clinical interpretation of viral hepatitis serology results

Clinical and laboratory triggers for referral	Further testing & actions	
	Cardiologist	ID specialist
BP control - Blood pressure (BP) $\geq 130/80$ mmHg - High CV risk - Abnormal ECG/CV physical exam	- Insist on weight management (BMI target in the normal range of 20 to 25 kg/m²) and smoking cessation - Initiate treatment with an ACEi or ARB plus CCB - Repeat ECG every 6 months or at least yearly Special attention: - Check DDIs with beta-blockers and calcium channel blockers in patients on boosted ART regimens - Exercise caution with combinations of multiple antihypertensive medications in people with history of orthostatic hypotension	- Insist on weight management (BMI target in the normal range of 20 to 25 kg/m²) and smoking cessation - Refer patient to cardiologist - Recommend echocardiography and/or NT-proBNP testing if suspicion of (worsening) CV disease
Lipid management - Abnormal lipid profile or - CV disease risk based on SCORE2 assessment	- Initiate pitavastatin. If not available: initiate atorvastatin or rosuvastatin in low initial doses - If LDL-c target is not reached: associate bempedoic acid alone or in combination with ezetimibe Special attention: - Simvastatin is contraindicated in patients on boosted ART regimens	Refer to cardiologist if LDL-c ≥ 70 mg/dL (or ≥ 55 mg/dL in very high-risk patients and in secondary prevention)
Pulmonary hypertension - Consult the pneumologist - Ensure interdisciplinary management	- Treatment includes prostacyclin analogues (e.g. epoprostenol, iloprost) Special attention: - Check DDIs with endothelin receptor antagonists and PDE5 inhibitors for patients on boosted ART regimens	People living with HIV are at high risk of pulmonary hypertension. In case of clinical suspicion (unexplained dyspnea, fatigue, signs of right heart strain), refer the patient to cardiology and pneumology for confirmation and treatment

ACEi – angiotensin converting enzyme inhibitor; ARB – angiotensin receptor blocker; ART – antiretroviral therapy; BMI – body mass index; BP – blood pressure; CCB – calcium channel blocker; CV – cardiovascular; DDIs – drug-drug interactions; ECG – electrocardiogram; HIV – human immunodeficiency virus; LDL-c – low-density lipoprotein cholesterol; NT-proBNP – N-terminal pro-B-type natriuretic peptide; PDE5 – phosphodiesterase type 5; SCORE2 – Systematic COronary Risk Evaluation 2.

Figure 3. Recommendations for the management of CV risk and/or CV disease in people living with HIV^{7,18,19}

strongly correlated to chronic inflammation, as well as to a cumulus of traditional risk factors, such as hypertension, diabetes mellitus, or smoking. ART has significantly reduced the incidence of classic HIV-associated nephropathy. However, other forms of CKD have become increasingly prevalent, including chronic kidney injury related to aging-associated comorbidities such as high blood pressure and diabetes.^{20,21} Additionally, hepatitis B virus (HBV) and hepatitis C virus (HCV) co-infections have been associated with an increased risk of CKD progression, potentially mediated by immune-

related mechanisms such as membranoproliferative glomerulonephritis.²¹⁻²³

Furthermore, tenofovir disoproxil fumarate (TDF)-containing ART regimens can cause kidney disease through proximal tubulopathy, with risk increased by pre-existing renal disease, older age, lower body weight, and boosted protease inhibitor (bPI) use. In patients with documented tubular proteinuria or glucosuria and in patients with rapidly declining kidney function, a switch to a tenofovir alafenamide (TAF)-containing regimen or a tenofovir-sparing regimen can be considered.⁷

Regular monitoring of renal function through estimated glomerular filtration rate (eGFR), phosphate levels and urinalysis is recommended in all people living with HIV to ensure early detection and appropriate management of kidney disease (Table 1). Particular attention should be paid to the potential nephrotoxic effects of ART and other co-administered medications, especially in the presence of comorbidities. Beyond controlling conventional risk factors, an individualized approach should include a thorough review of all medications and substances used, in order to identify and minimize nephrotoxic interactions. Proteinuria and decreasing eGFR are important signs of CKD progression and should prompt referral to specialized nephrology care.²¹ Figure 4 summarizes the clinical scenarios when referral to nephrology is indicated, with the need for enhanced communication between nephrologists and ID specialists.

Metabolic disease

Weight gain and metabolic complications are increasingly recognized among people living with HIV, reflecting a changing clinical profile in the context of long-term ART and others factors such as ageing, sedentary lifestyle, and dietary habits.^{26,27} Studies suggest that the prevalence of obesity and diabetes in patients living with HIV is now similar to or even higher than that in the general population.^{16,27,28} Notably, diabetes may occur at younger ages and even in the absence of obesity.²⁷ Identified risk factors for insulin resistance and type 2 diabetes include older age, high BMI, increased waist circumference, physical inactivity, poor diet, and family history.¹⁶ Moreover, metabolic dysfunction-associated steatotic liver disease (MASLD) in people living with HIV is emerging as a relevant comorbidity, often linked to male sex, obesity, diabetes, and associated with hepatic steatosis.¹⁶ However, the role of contemporary ART on MASLD remains underexplored.⁷

Similar to non-HIV specific management, healthcare professionals are reminded to provide lifestyle advice at each opportunity, encouraging patients to follow the usual guidance for weight control, healthy diet, physical activity and

smoking/substance use cessation (Figure 5). As part of routine screening, body weight, BMI, and waist circumference should be assessed at least annually.^{7,29} More frequent monitoring may be appropriate at each clinical visit in individuals with known diabetes, obesity, or other metabolic or cardiovascular risk factors.²⁹

Fasting plasma glucose is preferred in the screening for diabetes in people living with HIV, since HbA1c values tend to underestimate glucose levels in this population.³⁰ Diagnosis of HIV infection does not change the blood glucose targets in patients with diabetes mellitus (fasting and preprandial capillary plasma glucose 80-130 mg/dL, postprandial capillary plasma glucose <180 mg/dL and/or HbA1c target <7%).³¹ Treatment should include one novel class with proven cardio-reno-metabolic efficacy (SGLT2 inhibitors or GLP-1 receptor agonists). These two classes of medication have demonstrated efficacy in HbA1c reduction, positive impact on weight, significant reduction of cardiovascular and renal adverse outcomes (major adverse CV events, heart failure, CKD) without increased risk of hypoglycemia.^{7,32}

Bone disorders

Osteopenia and osteoporosis are more prevalent in people living with HIV compared to the general population, with a higher risk of low bone mineral density (BMD) and fractures.^{34,35} In addition to the traditional risk factors (female sex, age, low vitamin D levels, low sex hormone levels), ART regimens containing TDF or a bPI may further contribute to BMD reduction in people living with HIV.^{7,16}

Given the increased risk of BMD loss and fragility fractures in people living with HIV, clinicians should implement early screening strategies, monitor bone-related biomarkers, assess fracture risk using validated tools, and manage modifiable risk factors in collaboration with endocrinologists (Figure 6).

Preventing vertical transmission of HIV

Worldwide, almost half of all people living with HIV are women and girls.¹ A similar phenomenon is also seen in Romania, where women represent 36.5% of the overall patient

Table 1. Frequency of renal function monitoring and assessment in people living with HIV^{7,24,25}

Clinical situation	Serum creatinine (eGFR ^a)	Complete urinalysis ^b
At ART initiation	Yes	Yes
No renal risk, not on TDF	Every 6-12 months	Annually
On TDF	Every visit	Every visit
eGFR <60 mL/min/1.73 m ²	Every 3 months or more often	Every 3 months or more often
Comorbidities (HBP, diabetes)	Every 6 months	Every 6-12 months

^aFor eGFR calculation, the CKD-EPI formula is recommended. Use online calculators, as available.

^bIncluding proteinuria, hematuria, glucosuria and urine protein/creatinine ratio in case of proteinuria.

ART – antiretroviral therapy; **eGFR** – estimated glomerular filtration rate; **HBP** – high blood pressure; **TDF** – tenofovir disoproxil fumarate.

Clinical and laboratory triggers for referral	Further testing & actions	
	Nephrologist	ID specialist
Proximal tubulopathy - Progressive reduction of eGFR (≤ 90 mL/min) with no clear cause - Low serum phosphate and/or high UP/C - CKD (decreasing/stable eGFR ≤ 60 mL/min) - Tubular proteinuria	Check (on a case-by-case basis): - Serum and urinary phosphate levels - Serum and urinary glucose - Serum HCO_3^- and urinary pH - Serum and urinary uric acid - Serum and urinary K^+	- Replace TDF with TAF or tenofovir-sparing regimen - If hypophosphatemia, consider renal bone disease and refer to endocrinologist for osteoporosis assessment
Nephrolithiasis - Crystalluria - Erythrocytes and leukocytes in urine - Back pain - AKI	- Enhanced urinary tests (assessment of crystals) - Exclude other causes of nephrolithiasis - Imaging of the renal tract	- If on ATV, switch to another regimen - Refer to urologist and nephrologist, if needed
Interstitial nephritis - Progressive reduction of eGFR - Tubular proteinuria or hematuria - Presence of eosinophils in the urine - Leukocyte casts	Imaging of the renal tract to rule out obstruction or structural abnormalities	If on ATV, switch to another regimen
HIV-associated nephropathy - Nephrotic-range proteinuria - Rapidly declining eGFR $\pm$ edema - Suspicion especially at: CD4 <200 cells/mm³, HIV viremia and nonadherence to ART $\pm$ black ethnicity	- Laboratory testing, including renal biopsy - Initiate ACEi or ARBs to manage proteinuria and hypertension - Kidney replacement therapy with dialysis or transplantation, if needed	- Rapid ART initiation if not on ART - Switch TDF to TAF or a tenofovir-sparing regimen - If on ATV, switch to another regimen
Progressive reduction of eGFR with no additional trigger	- Complete assessment for CKD - Complete assessment for proximal tubulopathy, including UP/C and UA/C - Imaging of the renal tract	If on TDF and/or ATV, switch to another ART regimen

ACEi – angiotensin converting enzyme inhibitor; **AKI** – acute kidney injury; **ARB** – angiotensin receptor blocker; **ART** – antiretroviral therapy; **ATV** – atazanavir; **CKD** – chronic kidney disease; **eGFR** – estimated glomerular filtration rate; HCO_3^- – bicarbonate; K^+ – potassium; **TAF** – tenofovir alafenamide; **TDF** – tenofovir disoproxil fumarate; **UA/C** – urine albumin/creatinine ratio; **UP/C** – urine protein/creatinine ratio.

Figure 4. Recommendations for the management of renal risk factors and kidney disease in people living with HIV^{7,24,25}

Clinical and laboratory triggers for referral	Further testing & actions	
	Diabetologist	ID specialist
BMI control - Sedentary lifestyle - Dietary habits	- Provide lifestyle advice at each opportunity: guidance for weight control, healthy diet, physical activity and smoking/substance use cessation - Assess body weight, BMI, and waist circumference annually or more frequently in case of metabolic or cardiovascular risk factors - Some ART agents may be associated with weight gain (INSTIs or TAF), weight suppression (TDF), or metabolic risk (older PIs) - Consider "return to health" weight gain in patients starting ART; weight gain may level off over time - Unclear impact of switching ART for weight management.	
Glucose levels management	FPG screening is preferred over HbA1c; HbA1c values tend to underestimate glucose levels in people living with HIV	- Perform FPG test before ART initiation or switch, and 3-6 months afterwards. If initial screening results are normal, FPG can be checked annually - If FPG exceeds the normal range, refer the patient to the diabetologist for further evaluation
Diabetes Consider diabetes if any of the following are met: - Fasting plasma glucose (FPG) ≥ 126 mg/dL - HbA1c $\geq 6.5\%$ - Random plasma glucose ≥ 200 mg/dL with hyperglycemic symptoms 2-hour plasma glucose ≥ 200 mg/dL during an OGTT Consider prediabetes if: - Impaired fasting glucose (FPG = 100-125 mg/dL) - Impaired glucose tolerance (2-hour plasma glucose in OGTT between 140-199 mg/dL) - HbA1c between 5.7% and 6.4%	- Initiate or adjust antidiabetic therapy; consider DDIs - Consider SGLT2 inhibitors or GLP-1 receptor agonists in patients with type 2 diabetes and additional CV or renal risk - Assess for insulin resistance, metabolic syndrome, and MASLD - Initiate lifestyle interventions (diet, physical activity, weight control) - Monitor renal function and glycemic control (HbA1c, FPG) - Screen for diabetic complications (retinopathy, nephropathy, neuropathy)	- Be aware of increased diabetes risk even in non-obese or younger people with HIV - Monitor for glycemic control at each visit and reinforce adherence to both ART and metabolic treatment plans - Monitor potential ART-associated metabolic changes - Collaborate with the diabetologist to appropriately manage potential DDIs

ART – antiretroviral therapy; BMI – body mass index; CV – cardiovascular; DDIs – drug-drug interaction; FPG – fasting plasma glucose; GLP-1 receptor agonist – glucagon-like peptide 1 receptor agonist; HbA1c – hemoglobin A1c; INSTIs – integrase strand transfer inhibitors; MASLD – metabolic dysfunction-associated steatotic liver disease; OGTT – oral glucose tolerance test; PIs – protease inhibitors; SGLT2 – sodium-glucose transporter 2 inhibitor; TAF – tenofovir alafenamide; TDF – tenofovir disoproxil fumarate.

Figure 5. Recommendations for the management of weight and diabetes in people living with HIV^{7, 29-33}

Further testing & actions	
Endocrinologist	ID specialist
- Assess BMD by DXA in people living with HIV and risk factors - If normal BMD, repeat DXA every 3-5 years - Monitor serum calcium, phosphate, alkaline phosphatase, and 25(OH) vitamin D levels - Evaluate for secondary causes of osteoporosis (hypogonadism, renal phosphate wasting, chronic glucocorticoid treatment, etc.) - Recommend calcium and vitamin D supplementation where deficiency or insufficiency is confirmed - Consider antiresorptive therapy based on BMD results and fracture risk assessment - Provide lifestyle counseling (smoking and alcohol cessation, weight-bearing physical activity, adequate calcium intake)	- Screen all people living with HIV for bone health, particularly postmenopausal women, men ≥ 50 years, or with other risk factors - Evaluate fracture risk using FRAX, every 2 years starting at age 40 - Request DXA in patients with high fracture risk on FRAX, low BMI, previous fractures, alcohol excess, chronic glucocorticoid treatment, or ART regimens with known bone toxicity (TDF or bPI) - Consider switching from TDF to TAF or a tenofovir-sparing regimen in high-risk patients - Consider switching away from bPI, whenever possible - Monitor and correct vitamin D deficiency - Collaborate with endocrinology for management of severe osteopenia, osteoporosis, or recurrent fractures

ART – antiretroviral therapy; BMD – bone mineral density; BMI – body mass index; bPI – boosted protease inhibitor; DXA – dual energy X-ray absorptiometry; FRAX – fracture risk assessment tool; TAF – tenofovir alafenamide; TDF – tenofovir disoproxil fumarate.

Figure 6. Recommendations for the management of bone health in people living with HIV^{7,16,36-38}

population, and 36.4% among persons of reproductive age.⁵ Romania is among the countries that offer universal unconditional access to ART. Pregnant women with HIV should be offered sufficient information in an adapted language, with the aim of ensuring adequate ART throughout and beyond pregnancy and to adopt an optimal strategy for birth and feeding of their newborn in order to minimize the risk of HIV vertical transmission.

HIV testing should be routinely performed in all women who are trying to conceive; testing should be performed by any healthcare practitioner that they consult, regardless of the medical specialty.

During all pregnancies, HIV testing should be performed preferably at the first prenatal visit, and again in the third trimester, or more frequently, if specific risk factors are present.³⁹ Importantly, testing of the partner(s) is also warranted, to ensure comprehensive care, and to prescribe PrEP where needed.⁴⁰

In women in whom HIV infection is diagnosed during pregnancy, rapid ART initiation is of utmost importance, whereas in those already diagnosed with HIV, it is essential to adapt the existing ART regimen to a pregnancy-adequate option. Important further considerations include screening for co-infections (HBV, HCV, STIs, SCORTCH⁴¹), evaluating co-morbidities and careful intrauterine ultrasound monitoring.

Vertical HIV transmission can occur during pregnancy, delivery, or breastfeeding. The maternal HIV viral load at the time of delivery is the strongest predictor of transmission risk, whereas sustained virological suppression throughout pregnancy can reduce the risk of vertical transmission to below 1%.⁴⁰ To minimize this risk as much as possible, obstetrics, neonatology and infectious disease practitioners should closely coordinate care.

In women with newly initiated or modified ART, HIV viral load should be measured at baseline, 2 weeks after starting or switching ART, then monthly until undetectable. Thereafter, HIV viral load should be checked every 2 months, including at 36 weeks and at delivery. A similar HIV viral load frequency is also

recommended for patients already on ART prior to pregnancy.⁷ More frequent monitoring may be considered in cases that use non-standard regimens or that display suboptimal adherence.^{7,42} CD4 count should be measured in the first trimester of pregnancy; additional monitoring is only required if baseline CD4 <350 cells/mm³ or if clinically indicated.⁴²

Romania continues to implement a zero-risk policy regarding vertical transmission. This is particularly important, as the principle of U=U (undetectable = untransmissible) has not been confirmed for vertical transmission.⁷ Official recommendations state that delivery should be performed by elective caesarean section, usually at 38 weeks of gestation, before the onset of labor or membranes rupture, even in patients who exhibit virological suppression. For women diagnosed with HIV during labor, caesarean section should be performed in all cases when medically possible, intravenous zidovudine should be administered during labor if available, and ART should be initiated promptly.⁷

Postnatal prophylaxis (PNP) with 2 antiretroviral (ARV) drugs should be administered to all newborns born to mothers living with HIV, starting immediately from the first hours after birth, and continued for 4 weeks. In cases considered to be at high-risk, 3 ARVs can be used, and PNP can be continued up to 6 weeks. Intensive HIV testing of the infant is still recommended in the country, through HIV viral load, serology and Western Blot, performed at birth, 2 weeks after stopping PNP, and subsequently at 4-6, 9, 12, 18 and 24 months of age. In cases considered to be at high risk of transmission, additional HIV viral load tests should be performed whenever indicated, on a case-by-case basis. If the HIV viral load test is ever detectable, immediate age-appropriate ART should be started. Newborns should be referred to the pediatric departments of regional HIV centers in the country, within the first 72 hours of life, and monitored according to national guidelines, with continued coordination in multidisciplinary teams composed of infectious disease, obstetrics, neonatology and primary care specialists.

Breastfeeding is not recommended in Romania, as it carries a residual risk of HIV transmission, including in mothers with good adherence to ART and with undetectable HIV viral load;⁴⁰ even low risks are not considered acceptable under the country's zero-risk policy. In cases where mothers do decide to breastfeed despite the formal recommendation for formula feeding, HIV viral load should be tested monthly in the mother and the infant; breastfeeding should be stopped in case of detectable viral load, and post-exposure prophylaxis immediately started in the infant.⁷

Opportunistic infections

With the introduction of ART, which has dramatically improved the survival and health of persons living with HIV, the incidence of many opportunistic infections (OIs) has decreased. However, OIs still occur in individuals with HIV, particularly in those who do not achieve optimal immune recovery.

Several factors contribute to the continued presence of OIs in the ART era. These include: late diagnosis and delayed ART initiation, lack of adherence to ART, HIV drug resistance, lack of OI prophylaxis in patients with severe immunosuppression, and suboptimal access to healthcare services.

HIV infection is associated with varying degrees of immunosuppression, especially due to the depletion of CD4 cells. In Romania 58% of all new cases reported in 2024 presented late, with a CD4 cell count <350 cells/mm³.⁵ Immunosuppression also occurs in cases of lack of adherence to ART or even ART interruption. Adherence can be challenging due to factors such as socioeconomic issues, stigma, mental health issues, and side effects of ART, especially with older generation ARVs. Poor adherence or abandoned treatment can lead to incomplete viral suppression or loss of viral suppression altogether, which in turn increase the risk of OIs.

Typically, when the CD4 cell count drops below 200 cells/mm³, people living with HIV are at increased risk of OIs. In many cases, an OI may be the first clinical manifestation of undiagnosed HIV. Frequently, patients initially present to medical specialties other than

infectious diseases, and may either be unaware of their HIV status or choose not to disclose it due to fear of stigma. Thus, a severe, persistent, or recurrent infection or an infection with a pathogen not typically associated with such severity should serve as a red flag for clinicians across specialties, to prompt HIV testing along with appropriate immunological assessment.

If HIV infection is diagnosed in the context of an OI, the patient should be promptly evaluated by an infectious disease specialist to determine the optimal timing for initiating ART. In the latest version (12.1) of the EACS guidelines, part 1, section 4, some updates on management and prevention of OIs and of other emergent infections were included.⁷ In most cases, ART initiation is generally recommended as soon as possible, no later than two weeks after starting treatment for the OI,⁷ although this may vary depending on the specific infection and clinical presentation. The two main exceptions are tuberculous meningitis and cryptococcal meningitis, where ART initiation should be deferred for approximately 4 weeks, but could be considered earlier in selected cases in high-resource settings where close monitoring and optimal treatment is available.⁷ The infectious disease specialist will determine the optimal timing individually, based on recommendations from guidelines and the most up-to-date evidence from randomized clinical trials.

In patients already known to be living with HIV, treating physicians should consider adjustment of the ART regimen to ensure virological suppression and to prevent potential drug-drug interactions with agents used for OI treatment.⁷

OIs remain leading causes of hospitalization, morbidity, and mortality in people living with HIV.^{43,45} Maintaining an undetectable HIV viral load and a high CD4 count is essential for preventing OIs.

Important considerations about the most frequent OIs seen in clinical practice in Romania are presented below.

Cryptococcal meningitis. In patients with CD4 cell counts <100 cells/mm³ there is a strong recommendation for serum cryptococcal antigen

(CrAg) screening, as key tool for reducing mortality for cryptococcal infection. If serum CrAg is positive, lumbar puncture is mandatory; if cryptococcal meningitis is diagnosed, then guideline-recommended antifungal treatment should be started. If the test for serum CrAg is positive and cryptococcal meningitis or cryptococcal infection of other sites are excluded, pre-emptive antifungal therapy with fluconazole should be started before initiating/reinitiating ART, to prevent the development of invasive cryptococcal disease.⁴⁶⁻⁴⁸

***Pneumocystis jirovecii* pneumonia.** This is among the most frequent OIs seen in patients with advanced HIV infection (CD4 cell counts <200 cells/mm³), but can also occur with other causes of immunosuppression, such as prolonged high-dose glucocorticoid treatment, or cancer chemotherapy. Pneumologists and infectious diseases physicians should consider this diagnosis in all patients with immunosuppression, persistent dry cough, and progressively worsening dyspnea.⁷

Esophageal candidiasis. Caused by *Candida albicans*, this is another common OI in people with HIV and low CD4 cell counts. This fungal infection leads to white patches on mucosal surfaces and can cause significant pain and difficulty swallowing, symptoms that should raise the clinical suspicion and prompt diagnosis in patients with immunosuppression.⁷

Cerebral toxoplasmosis. In the updated EACS guidelines, trimethoprim-sulfamethoxazole (TMP-SMX) has been added as an additional preferred regimen for the treatment of *Toxoplasma* encephalitis.⁷ This is particularly relevant for Romania, where other agents used for the treatment of toxoplasmosis are often not available. This new recommendation is based on a recent meta-analysis confirming effectiveness and safety of TMP-SMX for this indication.⁴⁹

Infection with *Mycobacterium tuberculosis*. Co-infection with *Mycobacterium tuberculosis* accelerates HIV disease progression and

complicates treatment regimens. The rise of multidrug-resistant and extensively drug-resistant tuberculosis emphasizes the need for novel diagnostics and antimicrobial agents, and collaboration with experts in resistant TB.⁵⁰ Concurrent TB meningitis is one of the few indications for delaying ART initiation by 4 weeks; in case the CD4 cell count is very low (<50 cells/mm³), ART can be initiated sooner, within the first 2 weeks after starting TB treatment, only if close monitoring can be ensured. Corticosteroids are usually recommended as adjuvant treatment. Results from a recent clinical trial did not show a clear clinical benefit of corticosteroids in the management of TB meningitis in people living with HIV but these data were insufficient to modify the existing recommendation.⁵¹

Infection with nontuberculous mycobacteria (NTM). NTM have emerged as significant pathogens in people living with HIV, particularly those with advanced immunosuppression. Infections with NTM present diagnostic challenges due to their insidious onset and variable clinical presentations, and they often require prolonged and complex treatment regimens.⁷

Cytomegalovirus (CMV) infection. CMV can cause significant morbidity in people living with HIV with severe immunosuppression. CMV retinitis is the most common manifestation, leading to vision loss if untreated, but CMV can also affect other organs, including the gastrointestinal tract, lungs, and the central nervous system. Antiviral treatment should be started promptly, and followed by secondary prophylaxis for at least 3 months or until improvement of the immune function (CD4 cell count >100 cells/mm³) and control of HIV infection are obtained, whichever is longer.⁷

Progressive multifocal leukoencephalopathy (PML). This is a severe, progressive, demyelinating disease of the central nervous system caused by JC polyomavirus in immunocompromised individuals. In people living with HIV and advanced

immunosuppression PML presents with multifocal neurological deficits that often include cognitive decline, ataxia, motor dysfunctions, visual impairment. Despite ART improving survival, PML remains associated with high morbidity and mortality.

Prophylaxis and preemptive therapy play critical roles in preventing many common OIs. For example, the use of TMP-SMX for preventing *Pneumocystis jirovecii* pneumonia and cerebral toxoplasmosis in patients with CD4 cell counts <200 cells/mm³, or the use of isoniazid for the prevention of tuberculosis in high-risk individuals, are widely used prophylactic strategies in people living HIV with severe immunosuppression.⁷ Pre-emptive antifungal therapy with fluconazole for cryptococcal disease has shown efficacy in reducing morbidity and mortality in high-risk populations.⁷ NTM prophylaxis with azithromycin, clarithromycin or rifabutin is considered for persons with CD4 cell counts <50 cells/mm³ who remain viremic on maximal ART, due to HIV drug resistance.⁷ The duration of prophylaxis depends on the improvement of the immune status, assessed through repeated CD4 cell count assessments.

Overall, effective prevention and management strategies, including early diagnosis, prophylactic treatments, ART optimization, and addressing adherence barriers, are essential in reducing the burden of OIs in the modern era of HIV care. Comprehensive care that integrates both biomedical and psychosocial interventions remains key to improving outcomes for individuals living with HIV.

Vaccination in people living with HIV

Vaccination is a key preventive measure for people living with HIV and a very important component of comprehensive care. Romania provides fully reimbursed access to a wide range of vaccines for people living with HIV.⁵² Reimbursed vaccine prescriptions can be issued by any physician, from any medical specialty. For this reason, it is essential that healthcare providers caring for people living with HIV prioritize an evaluation of the immunization status during their consultation, in order to subsequently offer appropriate recommendations

for vaccination. This is particularly important given that Romania continues to report high incidence rates of vaccine-preventable diseases, including measles, pertussis, and pneumococcal infections.

Table 2 summarizes key vaccination recommendations for people living with HIV in Romania. It is important to prioritize vaccination in people living with HIV, including patients with immunosuppression, as they are at the highest risk of severe outcomes if they acquire a vaccine-preventable disease. When making recommendations regarding vaccination, two important considerations should be taken into account: safety and efficacy.

Most vaccines can be safely administered anytime during the course of HIV infection, with contraindications only for live-attenuated vaccines (measles-mumps-rubella (MMR), chickenpox, live-attenuated influenza vaccine), which should be temporarily postponed in patients with immunosuppression (defined as CD4 count <200 cells/mm³ or CD4 $<15\%$, or presence of acquired immunodeficiency syndrome (AIDS)-defining conditions).^{7, 53} Even in these cases, vaccination should be considered as soon as the immune status improves, and thus it is not a matter of contraindicating the vaccine, but rather of identifying the optimal timing to allow safe administration of live-attenuated vaccines.

The second consideration is that of efficacy, and for patients living with HIV the best immunogenicity is seen for vaccines administered when there is sufficient immune competence, defined by a satisfactory CD4 count and virological suppression. Generally, accelerated vaccination schedules should be avoided. Since the aim is to protect patients living with HIV from vaccine-preventable diseases, most vaccines (see above) can and should be safely administered, even during immunosuppression. When possible, post-vaccination antibody titers should be assessed, and doses that had been administered during periods of suboptimal immune status or viral control may be repeated after immune recovery, to ensure an appropriate immune response (Figure 7).⁷

Table 2. Vaccination recommendations for people living with HIV in Romania

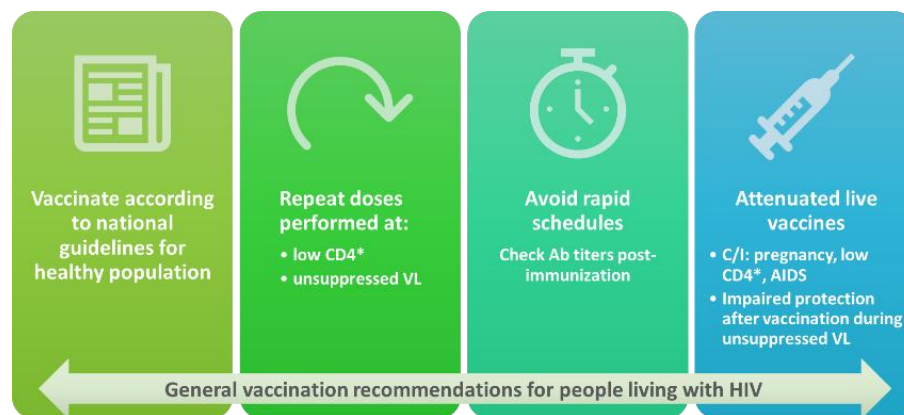
Vaccine	Notes
Hepatitis B vaccine	• Similar indication as for the general population: Birth dose for all newborns, followed by 3 more doses of hexavalent vaccine at 2, 4 and 11 months of age.⁵⁴ • If mother has positive HBsAg: antiviral prophylaxis during pregnancy; HBV vaccine birth dose + HBIG in newborn • For those not vaccinated at birth: 3-dose schedule (administered at 0, 1, 6 months).^{7,54} In case of non-response, the 3-dose regimen will be repeated; double doses of an age-appropriate vaccine can be used. • Promising data for the CpG-adjuvanted HepB vaccine,^{55,56} but not yet available in Romania.
Hexavalent vaccine (DTaP/IPV/Hib/HepB)	• Similar indication as for the general population: 3 doses for all infants with HIV exposure or infection, preferably at 2, 4 and 11 months old, identical to infants from the general population.⁵⁴
DTaP (±IPV)	• Similar indication as for the general population: At 6 years and 14 years, for all children and adolescents living with HIV. • Subsequently, Tdap (or Tdap+IPV, if available) boosters every 10 years and during each pregnancy.^{53,57} • For management of tetanus-prone wounds, any tetanus toxoid-containing vaccine can be safely used in people living with HIV
Pneumococcal vaccine	• Similar indication as for the general population: for infants, 3 doses of PCV at the age of 2, 4 and 11 months, administered concomitantly with the hexavalent vaccine, at different anatomical sites. • PCV booster dose in the second year of life for children living with HIV. • Children living with HIV aged 1-5 years who were not vaccinated during infancy should receive at least two doses, administered at least 8 weeks apart.⁵⁴ • For all people living with HIV with ages above 5 years, administer one dose of PCV; can be followed by PPSV-23 after ≥8 weeks, or a broader-coverage PCV, if the initial vaccination was performed with PCV-13 or PCV-15.⁷ • Pneumococcal vaccines are reimbursed for patients living with HIV in Romania, regardless of age.
Meningococcal A, C, W-135, Y conjugate vaccine (MenACWY)	• Infants: multi-dose regimen depending on age at first dose.⁵⁸ • ≥1 year old: 2-dose primary regimen (at least 8 weeks apart), followed by boosters every 5 years.⁵⁹
Meningococcal B vaccine (MenB)	• 4CMenB: ○ Infants and children up to 2 years old: multi-dose regimen depending on age at first dose. ○ Age ≥2 years old: 2-dose primary regimen (at least 8 weeks apart); one booster dose 5 years after completing primary vaccination.⁵⁹ • MenB-FHbp: ○ Age ≥10 years old: 2-dose primary regimen (at least 6 months apart); consider boosters. • The MenB vaccines currently available in Romania cannot be used interchangeably; the same product should be used for primary vaccination and booster doses. • Meningococcal vaccines are reimbursed for patients living with HIV in Romania, regardless of age.

HPV vaccine	• Similar indication as for the general population • Indicated for all females and males; 9-valent vaccine will be used.^{7,53} ◦ Age 9-14 years old (at first dose): 2-dose series (6 months apart). ◦ Age ≥15 years old (at first dose): 3-dose series at 0, 2, 6 months. • If previously vaccinated with a 2-valent or 4-valent vaccine: repeat primary immunization with 9-valent vaccine. • No need to test for HPV before vaccinating. • Can be administered without delay if positive for one or more HPV genotypes; will protect against the other HPV genotypes. • Can be administered without delay in the presence of intraepithelial lesions; will protect against future infection and disease from new HPV genotypes. Specialized gynecological treatment for intraepithelial lesions should also be provided. • Check upper age limit for reimbursement; no upper age limit for medical recommendation – continue to recommend primary vaccination at any age if sexually active.
Influenza vaccine	• Everyone aged ≥6 months old: vaccinate annually, with inactivated influenza vaccine.^{7,53} • Age ≥60 years old: prioritize use of high-dose inactivated influenza vaccine.⁶⁰ • Avoid live-attenuated influenza vaccine in people living with HIV.⁵³
Hepatitis A vaccine	• Recommended for all people living with HIV aged ≥1 year old.⁷ • Monovalent vaccine is preferred, 2 dose series (6 months apart).^{7, 53} • Check antibody response (HAV IgG) in case of immunosuppression. In case of non-response following the 2-dose series, administer one more dose and recheck HAV IgG one month later.
COVID-19 vaccine	• All people living with HIV should be vaccinated against COVID-19 with an age-appropriate primary series followed by variant-adapted boosters based on local SARS-CoV-2 epidemiology.⁷
RSV vaccine and monoclonal antibodies	• RSV vaccines: ◦ Similar indication as for the general population. ◦ RSVPreF: Pregnant women (gestational weeks 24-36) and patients aged ≥60 years old: one dose.^{53,61} ◦ RSVPreF3: Patients aged ≥60 years old: one dose. • RSV monoclonal antibodies: ◦ Similar indication as for the general population: infants exposed to or living with HIV can receive monoclonal antibodies for RSV prophylaxis. ◦ This includes infants born to mothers living with HIV who received RSV vaccine during pregnancy.⁶² ◦ Long-acting formulations available (single dose per season).^{63,64} ◦ If born during the RSV season, administer long-acting monoclonal antibody dose shortly after birth. If born outside of the RSV season, administer the dose at the beginning of the first RSV season. ◦ Administer a second dose of a long-acting monoclonal antibody (nirsevimab) to children who are 8-19 months old at the beginning of the second RSV season.⁶⁵
Rotavirus vaccine	• All children, under 6 months, with HIV exposure or HIV infection.⁶⁶⁻⁶⁸ • Oral administration. • Monovalent rotavirus vaccine: 2-dose regimen (4 weeks apart) starting at 6 weeks old; administer second dose ideally before 16 weeks of age, not later than 24 weeks. • Pentavalent rotavirus vaccine: 3-dose regimen (4 weeks apart) starting at 6 weeks old; administer third dose ideally before 20 weeks of age, not later than 26 weeks.

Rabies vaccine	• If pre-exposure rabies vaccination is recommended: ○ For CD4 ≥ 200 cells/mm³: 2-dose intramuscular schedule (days 0 and 7). ○ For CD4 < 200 cells/mm³ or detectable viremia: 3-dose regimen (days 0, 7, and 21 or 28).⁷ • If post-exposure prophylaxis (PEP) is recommended, perform wound cleaning, consider HRIG based on risk evaluation, plus vaccine: ○ Previously unvaccinated individuals: ▪ For CD4 ≥ 200 cells/mm³, 4-dose schedule (days 0, 3, 7, 14). ▪ For CD4 < 200 cells/mm³ or detectable viremia: 5-dose schedule (days 0, 3, 7, 14, 28) with HRIG.⁷ ○ Previously vaccinated individuals: ▪ For CD4 ≥ 200 cells/mm³, 2-dose schedule (days 0, 3-7).⁶⁹ ▪ For CD4 < 200 cells/mm³ or detectable viremia: 5-dose schedule (days 0, 3, 7, 14, 28).⁷
Typhoid fever vaccine (polysaccharide formulation available)	• Polysaccharide vaccines generally exhibit lower immunogenicity in people living with HIV, especially in the presence of immunosuppression.⁷ • The vaccine can be administered prior to travel to endemic areas in people aged ≥ 2 years old. • One dose administered at least 2 weeks prior to departure. • Strict food and water precautions remain critical components of prevention and should also be applied, as immunogenicity might be suboptimal.
Live vaccines – assess the right timing for vaccination based on immune status (safe to administer at CD4 ≥ 200 cells/mm³)	
BCG vaccine	• Romania implements a single dose universal BCG vaccination at birth. • For infants living with HIV, BCG will be delayed until they are on ART and immunologically stable (i.e., CD4 $> 25\%$).^{54,70} • For newborns with unknown HIV status, perinatally exposed to HIV, BCG will be administered if no clinical signs of HIV are present.^{54,70} • Temporary contraindication: birth weight < 2500 g.
MMR vaccine	• In Romania, a 2-dose MMR vaccine schedule is indicated for all children (at age 1 and 5 years old), with a supplementary dose at age 9 months during measles epidemics. • If not vaccinated according to the National Immunization Program: 2 MMR doses administered at least 4 weeks apart.⁵³ • No need to check MMR serologies prior to vaccination. • Regardless of serology results, the vaccine is still indicated if IgG negative for any of the MMR components. • Temporary contraindications: ○ CD4 $< 15\%$ or CD4 < 200 cells/mm³. ○ Marked immunosuppression of a different cause. ○ Pregnancy; avoid pregnancy for one month after MMR.^{53,71} • Assess the right timing for MMR vaccination in people living with HIV: ○ Age ≤ 5 years: administer MMR if CD4 $> 15\%$ for ≥ 6 months, regardless of absolute CD4 count.⁷² ○ Age > 5 years: administer MMR if CD4 $\geq 15\%$ and CD4 ≥ 200 cells/mm³ for ≥ 6 months.⁵³
Varicella vaccine	• After age 1 year: 2 doses administered at least 6 weeks apart (3 months apart if first dose given earlier, between 9-11 months old).⁵³ • No need to check VZV serologies prior to vaccination, in children. • See MMR above for: <i>Temporary contraindications</i> and <i>Assess the right timing</i>
Yellow fever	• For travelers to/from endemic areas: administer 1 dose, if not contraindicated.^{7,63} • See MMR above for: <i>Temporary contraindications</i> and <i>Assess the right timing</i>

Mpox vaccine (live, non-replicating)	• Recommended in Romania for PEP only.⁷³ • Administer first dose as soon as possible following exposure; administer second dose 28 days later.
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ART – antiretroviral therapy; **BCG** – Bacille Calmette-Guérin; **COVID-19** – coronavirus disease 2019; **DTaP** – diphtheria, tetanus, and acellular pertussis vaccine; **HAV IgG** – hepatitis A virus immunoglobulin G; **HBV** – hepatitis B virus; **HepB** – hepatitis B vaccine; **Hib** – *Haemophilus influenzae* type b vaccine; **HIV** – human immunodeficiency virus; **HPV** – human papillomavirus; **HBIG** – hepatitis B immune globulins; **HBsAg** – hepatitis B surface antigen; **HRIG** – human rabies immunoglobulin; **IPV** – inactivated poliovirus vaccine; **MenACWY** – meningococcal conjugate vaccine against serogroups A, C, W-135, and Y; **MenB** – meningococcal B vaccine; **MenB-FHbp** – meningococcal serogroup B vaccine (factor H binding protein); **4CMenB** – four-component meningococcal serogroup B vaccine; **MMR** – measles, mumps, rubella; **PEP** – post-exposure prophylaxis; **PCV** – pneumococcal conjugate vaccine; **PPSV** – pneumococcal polysaccharide vaccine; **RSV** – respiratory syncytial virus; **RSVPreF/RSVPreF3** – RSV prefusion F protein-based vaccines; **SARS-CoV-2** – severe acute respiratory syndrome coronavirus 2; **Tdap** – tetanus, diphtheria, and acellular pertussis booster vaccine; **VZV** – varicella zoster virus.



*Low CD4 count is defined as CD4 <200 cells/mm³ or <15%

Ab – antibodies; AIDS – acquired immunodeficiency syndrome; C/I – contraindicated; VL – viral load.

Figure 7. General principles of vaccination for people living with HIV

Quality of life

Psychological factors play an important role in the life of patients with HIV, the defining element being “living with the diagnosis”. Emotional, affective and cognitive disorders, anxiety and depression are increasingly present, especially in young adults with old infection.

HIV infection has become a “chronic” disease, and for people living with HIV, this term may also be perceived as “progressive” and “not curable”, resulting in a specific emotional frailty expressed in many day-to-day interactions, including adherence to ART.⁷⁴

It is essential that the HIV care team includes a clinical psychologist, to provide ongoing counseling and mental health support. Ideally, psychological evaluation and support should be offered at every clinical visit, or more frequently

if the patient’s condition requires it. It is recommended to screen for anxiety and depression with two simple tools (PHQ-2 and GAD-2 or GAD-7 questionnaires), which may be applied by any healthcare professional interacting with people living with HIV; screening results can help identify patients who should be referred to the psychologist and/or psychiatrist for appropriate management.

In recent years, the evaluation of healthcare-related quality of life in people living with HIV has increasingly included standardized tools classified as patient-reported outcome measures (PROMs).⁷⁵ These instruments promote patient-centered care and capture valuable insights into patients’ perceptions and experiences related to HIV care, communication with healthcare teams, emotional health, stigma, overall well-being and

other lifestyle aspects.^{75,76} PROMs are typically self-completed questionnaires, and should ideally be used annually right before or during an assessment visit at the clinic, to facilitate an open conversation with the treating physician.⁷ In Romania, the National Institute for Infectious Diseases “Prof. Dr. Matei Balș” has facilitated the uptake of PROMs throughout the Romanian network.

A significant challenge is the continued presence of hidden stigma, including among healthcare professionals.⁶ In Romania, HIV-related stigma, either enacted or anticipated, is a persistent reality that may impact access to appropriate and non-discriminatory care.⁶ Thus, in addition to other measures, educational strategies for healthcare providers of all specialties are needed to reduce stigma and ensure equitable, respectful and patient-centered care for all people living with HIV.

Conclusions

The care of people living with HIV continues to evolve in response to advances in treatment and increase in life expectancy, with a focus on quality of life. For too long, the responsibility for HIV care rested solely with infectious disease specialists; today, it is increasingly clear that active involvement from all medical specialties is essential. The concept of multidisciplinary care now represents a standard of care approach, and frameworks for its effective implementation should be developed on all levels of care. International guidelines for HIV management are regularly updated, and their integration into everyday clinical practice should be prioritized, including within non-infectious disease specialties. This national consensus provides a practical, context-specific adaptation of international recommendations to Romanian healthcare realities. Effective care requires more than clinical knowledge, it relies on structured collaboration, timely referrals, and a shared commitment to addressing comorbidities and age-related conditions through evidence-based strategies. By embracing clinical frameworks and fostering coordinated, interdisciplinary approaches, we can ensure that HIV care in

Romania is not only responsive and equitable, but also sustainable.

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